

REVERSIBLE WATERMARK-GUIDED DATA OBFUSCATION TO PREVENT EXPLOITATION BY UNAUTHORIZED MODELS

Anonymous authors

Paper under double-blind review

ABSTRACT

The widespread availability of “free” data on the Internet has been key to recent advances in deep learning. However, it also raises serious concerns about the unauthorized use of public data to train commercial models. This paper investigates a critical issue in the domain of data privacy and deep learning: *Can data be made unexploitable to unauthorized models while being recoverable for authorized models?* To address this issue, we propose Reversible Data Obfuscation, the first method that leverages watermark in a reversible manner to control data usability for target models. Specifically, we employ a gradient-based search (GBS) combined with constraint-guided obfuscation (CGO) strategy to select watermarkable tokens, generating watermarked obfuscated data that remains imperceptible to humans while preserving normal data utility. For unauthorized models, watermarked obfuscated data increases the discrepancy between predicted and ground-truth labels, thereby effectively degrading model performance. In contrast, authorized models can extract the watermark to recover the original high-quality data for effective training. Experimental results on classification tasks show that models fine-tuned on the watermarked obfuscated data experience severe performance degradation, with accuracy dropping below 50%. Moreover, the performance of unauthorized models degrades progressively with repeated embedding. The obfuscated data preserve high semantic similarity with the original data, and the embedded watermarks can be reliably extracted for ownership verification. Our work establishes an important step towards making data unexploitable to unauthorized models while remaining exploitable to authorized models.

1 INTRODUCTION

Recent advances in deep learning have been fueled by the widespread availability of large-scale public datasets, many of which are sourced from the Internet with minimal restrictions. However, recent reports reveal that personal data is used to train or fine-tune commercial models without permission of data owners (Zhang et al., 2018; Hou et al., 2024). Some owners refuse the use of their personal data for model training. However, unauthorized models may still exploit such data to train their models. In this case, verifying the legal ownership of training data remains a concern.

Unauthorized use of personal data for model training constitutes a clear violation of ownership. Natural language watermarking is an effective technique for ownership protection. Watermark information (Kamaruddin et al., 2018; Yang et al., 2022; Liu et al., 2024; Liu & Bu, 2024) is embedded into text via word substitution, which enables the text to carry identification information without degrading its quality. Although the previous watermarking scheme aimed to preserve the quality of the text as much as possible, it still had some impact on text quality. Therefore, reversible watermarking schemes (Khan et al., 2014; Dragoi & Coltuc, 2015; Xiang et al., 2024; 2025) have been proposed, which allow for both the extraction of watermark information and the recovery of the original text.

While such watermarking schemes enable data traceability, they are insufficient to prevent models from exploiting these data to improve performance. To prevent unauthorized model training, prior work has demonstrated the effectiveness of injecting perturbations into dataset to degrade model performance, thereby making the data unlearnable to models. Techniques such as unlearnable ex-

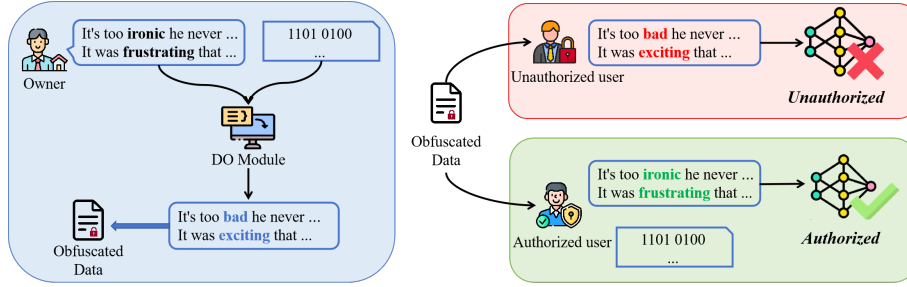


Figure 1: The framework of our data obfuscation (DO) method.

amples (Huang et al., 2021), adversarial attacks (Qi et al., 2024), and data poisoning (Qi et al., 2024) can effectively degrade the performance of deep learning models. However, they typically introduce permanent corruption to the dataset, rendering performance degradation of models trained on it.

In real-world scenarios, it is challenging to both authenticate data ownership and prevent unauthorized model training. To address this, we propose a *reversible watermark-guided data obfuscation* framework that allows data owners to embed model-sensitive perturbations into textual inputs, as illustrated in Figure 1. These perturbations, which are associated with owner-specific watermarks, degrade the performance of unauthorized models. Meanwhile, authorized users can accurately recover the original clean data and associated watermark information. Our method confuses unauthorized models while preserving utility for authorized ones. The data obfuscation (DO) module consists of Gradient-Based Search (GBS) and Constraint-Guided Obfuscation (CGO), facilitating the generalization of obfuscated data across target models. Models trained on recovered clean data achieve high performance, while those trained on obfuscated data suffer significant degradation. The embedded watermarks further support ownership verification.

Our key contributions are as follows:

- We propose a novel Reversible Data Obfuscation method that embeds a watermark-guided obfuscation into the data to degrade model performance. The watermark can be extracted for traceability and enables recovery of the original clean data.
- We propose a Constraint-Guided Obfuscation (CGO) method that leverages a Gradient-Based Search (GBS) to strategically identify the most influential tokens for watermark embedding, thereby optimizing the trade-off between obfuscation strength and data utility.
- We empirically demonstrate the effectiveness and feasibility of the proposed obfuscation method in significantly degrading the training performance of unauthorized models, while preserving the readability of the obfuscated data.

2 RELATED WORK

2.1 DATA PERTURBATION AND PROTECTION

Unlearnable examples aim to prevent use of personal data by making it ineffective for model training. Li & Liu (2023) propose error-minimizing noise to generate unlearnable text that misleads models into learning “nothing” from the perturbed data. Adversarial attacks (Qi et al., 2024) typically add imperceptible noise to input data at test time to mislead model predictions. Such attacks identify error-maximizing noise that significantly increases prediction error, and this perturbation can often be crafted as a universal pattern applicable across the entire test set (Moosavi-Dezfooli et al., 2017). In contrast, data poisoning attacks (Shan et al., 2024) aim to modify the training examples to degrade the model’s performance on clean examples. Du et al. (2024) leverages AI-generated poisoned text, obtained through continued writing or paraphrasing, to attack natural language processing models. Poisoned examples tend to cause degradation in model performance, but are often distinguishable from clean data, thereby compromising data utility (Carnerero-Cano et al., 2024).

To mitigate the risk of private information leakage from models, differential privacy (DP) has been widely adopted as a data protection strategy (Abadi et al., 2016; Gursoy et al., 2021; Dong et al.,

2022; Ponomareva et al., 2023; Dwork & Roth, 2014). By adding noise into the dataset, DP ensures individual-level privacy while still enabling data sharing. However, most DP-based schemes focus on balancing privacy and data utility (Wei et al., 2020; Pan et al., 2024; Yang et al., 2024; Shukla et al., 2025). They often suffer from degraded performance due to the difficulty in qualifying appropriate noise levels without knowledge of downstream tasks. While these methods are effective in degrading model performance, they generally lack the ability to restore the original clean examples, making it difficult to implement effective data usage control. Prior hiding-based unlearnable examples in vision Meng et al. (2024) introduce random noise during recovery and therefore cannot achieve lossless reconstruction.

2.2 REVERSIBLE WATERMARK

Liu et al. (2010) proposed a reversible text watermarking method that transforms selected words into integers, embedding them using an enhanced integer transform and difference expansion. Recently, Xiang et al. (2024) proposed a scheme for protecting sensitive textual information using a masked language model and prediction error expansion. Jiang et al. (2025) developed a source-aware reversible natural language watermarking method that proactively mines potential watermarkable positions and embeds the watermark via source-aware lexical substitution.

Building upon these efforts, we first apply a reversible watermarking method to generate obfuscated data, enabling ownership tracking and usage control, while intentionally constraining the performance of downstream models. The modifications introduced to create obfuscated data are minimal and imperceptible to humans, ensuring the obfuscated data remains natural and human-readable.

3 METHOD

3.1 OBJECTIVE OF DATA OBFUSCATION METHOD

We formulate the problem in the context of text classification. Considering the clean training datasets $\mathcal{D}_c = (x, y)_{i=1}^N$, where the i -th instance consists of a text x and its true label y for classification. We denote its obfuscated version by $\mathcal{D}_o = (x', y)_{i=1}^N$, where x' is the obfuscated version of training data $x \in \mathcal{D}_c$. Given test datasets $\mathcal{D}_t$, our goal is to transform the training datasets $\mathcal{D}_c$ into obfuscated datasets $\mathcal{D}_o$ such that model trained on the $\mathcal{D}_o$ will perform poorly on the test datasets $\mathcal{D}_t$.

Given a typical K -class classification task, a pretrained language model classifier f maps the input $x \in \mathcal{X}$ to the output label $y \in \mathcal{Y} = 1, \dots, K$. To generate obfuscated training data $\mathcal{D}_o$, we aim to maximize the classification loss with respect to the model f . Specifically, we use the standard cross-entropy loss $\mathcal{L}$ as the objective:

$$\arg \max_{x'} \mathcal{L}(f(x'), y) \quad (1)$$

By maximizing $\mathcal{L}$, we construct obfuscated examples that mislead the classifier and degrade its predictive performance.

3.2 GRADIENT-BASED SEARCH STRATEGY

To the best of our knowledge, we are the first to generate obfuscated data through a watermark-guided method. To identify the most influential tokens for watermark embedding, we compute token-level importance scores via a gradient-based strategy. Given an input token sequence $x = \{t_1, t_2, \dots, t_n\}$, we obtain its embedding representation $E = \{\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n\}$, where $\mathbf{e}_i \in \mathbb{R}^d$ is the embedding vector for token t_i . Let $P = \{1, 2, \dots, n\}$ denote the set of token positions in x .

The surrogate loss based on the ℓ_2 norm of the output logits is produced by a pretrained language model f , which is defined as:

$$\ell = \mathcal{L}(E) = \|f(E)\|_2 \quad (2)$$

where $f(E)$ denote output logits of model.

We compute the gradient of the loss with respect to each input embedding:

$$\nabla_{\mathbf{e}_i} \ell = \frac{\partial \ell}{\partial \mathbf{e}_i} \quad (3)$$

Algorithm 1 Gradient-based Search Strategy**Input:** A pretrained language model f , surrogate loss ℓ , and an input-label pair (x, y) **Output:** Optimal token position p

- 1: Generate $\nabla_{e_i} \ell$.
- 2: Generate importance scores S for all the tokens in the input-label pair (x, y) via first-order approximation.
- 3: Sort P in the descending order of S to obtain $\hat{P}$
- 4: **for** each token indexed $p \in \hat{P}$ **do**
- 5: **if** p satisfies all the constraints **then**
- 6: **return** p
- 7: **end if**
- 8: **end for**

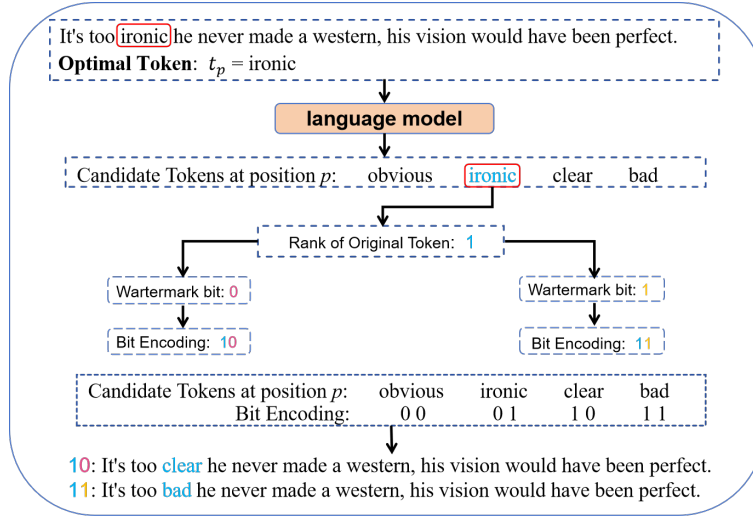


Figure 2: The procedure of watermark embedding for obfuscated data generation.

Then, the importance score s_i for each token t_i is calculated as the inner product between its embedding vector and the corresponding gradient:

$$s_i = \langle \nabla_{e_i} \ell, e_i \rangle \quad (4)$$

The score s_i measures the directional influence of the embedding on the loss, serving as a first-order approximation of the change in ℓ under a perturbation in the direction of e_i . Tokens with higher s_i are deemed more influential, and are prioritized for watermark embedding.

To generate obfuscated data that interferes with model training, we identify and modify the input token that has great influence on the model’s loss. Specifically, we compute an importance score for each token and rank them in descending order to guide token selection for modification. Let $\hat{P}$ denote the set of positions of the importance-ranked tokens.

Algorithm 1 outlines the procedure for selecting an optimal token t_p to be substituted within a given input-label pair (x, y) during a single iteration, with the goal of generating an obfuscated instance. The constraints of Step 5 in Algorithm 1 are specified as follows.

3.3 CONSTRAINT-GUIDED OBFUSCATION SEQUENCE GENERATION

To ensure both the reversibility of watermark-guided obfuscated data and its ineffectiveness for unauthorized model training, we introduce additional constraints to determine the final positions for watermark embedding during obfuscation.

Algorithm 2 Constraint-Guided Obfuscation Sequence Generation

Input: A pretrained language model f , token sequence $x = \{t_1, t_2, \dots, t_n\}$, target bit $b \in \{0, 1\}$, importance-ranked token positions $\hat{P}$

Output: Obfuscated token sequence x'_o

```

1: for each token indexed  $p \in \hat{P}$  do
2:   Feed  $x$  into language model  $f$  to obtain logits over vocabulary for token  $t_p$ 
3:   Obtain top four candidate tokens at position  $p$ :  $C = \{w_0, w_1, w_2, w_3\}$ 
4:   if  $t_p \in \{w_0, w_1\}$  then
5:     Encode first bit:  $b_1 \leftarrow 0$  if  $t_p = w_0$ , else 1
6:     Encode second bit:  $b_2 \leftarrow b$ 
7:     Determine substitution token  $t'_p \in C$  based on  $(b_1, b_2)$  encoding method
8:     Substitute  $t_p$  with  $t'_p$  to form  $x'_o$ 
9:     if  $\mathcal{L}[f(x'_o), y] > \mathcal{L}[f(x), y]$  and  $f(x'_o) \neq y$  then
10:      Append  $p$  to position list  $\mathbb{P}$ 
11:     return  $x'_o$ 
12:   end if
13: end if
14: end for

```

Algorithm 3 Reversible Bit Extraction and Text Recovery

Input: Obfuscated token sequence $x'_o = \{t'_1, t'_2, \dots, t'_n\}$, embedded positions $\mathbb{P} = \{p_1, p_2, \dots, p_m\}$, a pretrained language model f for token prediction

Output: Recovered bit string $b = \{b_1, \dots, b_m\}$, recovered text sequence $x = \hat{x} = \{t_1, \dots, t_n\}$

```

1: Initialize recovered sequence  $\hat{x} \leftarrow x'_o$ , bit string  $b \leftarrow []$ 
2: for  $p$  in reverse( $\mathbb{P}$ ) do
3:   Compute top four candidate tokens at position  $p$  via language model  $f$ 
4:   Obtain top four candidate tokens at position  $p$ :  $C = \{w_0, w_1, w_2, w_3\}$ 
5:   if  $t'_p = w_k$  then
6:     Recovered original token  $t_p = w_{k/2}$ 
7:     Recovered bit  $bit = k \bmod 2$ 
8:   else
9:     Flag  $t'_p$  as potentially tampered
10:  end if
11:  Append  $bit$  to recovered bit string  $b$ 
12:  Replace  $\hat{x}[p]$  with  $t_p$ 
13:  return  $b, \hat{x}$ 
14: end for

```

As illustrated in Figure 2, for each token indexed by $p \in \hat{P}$, we feed the context $x = \{t_1, t_2, \dots, t_p, \dots, t_n\}$ into the language model to obtain the logit distribution over the vocabulary corresponding to the token at position p . If the original token t_p is ranked among the top two candidate tokens, the position is deemed eligible for watermark embedding.

To embed two bits of information, we utilize the top four candidate tokens $C = \{w_0, w_1, w_2, w_3\}$ produced by the language model at position p , encoding both the rank of the original token and the target watermark bit. Specifically, the first bit encodes the relative rank of t_p among the top two candidate tokens: it is set to 0 if t_p is the top-ranked token, and 1 if it is ranked second. The second bit corresponds to the target watermark bit $b \in \{0, 1\}$ intended for embedding.

Given the rank of t_p among the top two candidate tokens and the target bit b , a substitution is performed to produce a potentially obfuscated sequence x'_o .

To ensure that the generated sequence is obfuscated with respect to the target classifier f , further constraints are imposed. The substitution is accepted only if the loss increases:

$$\mathcal{L}[f(x'_o), y] > \mathcal{L}[f(x), y], \quad (5)$$

and the predicted label changes:

$$f(x'_o) \neq y, \quad (6)$$

where y is the ground-truth label. If both conditions are satisfied, x'_o is retained as a successful obfuscation. We record the token index p into a position list $\mathbb{P}$, which is later used for watermark bit extraction. Otherwise, the algorithm proceeds to the next candidate token position $p \in \hat{P}$.

The complete procedure for evaluating constraint satisfaction at each position is described in Algorithm 2.

3.4 BIT EXTRACTION AND TEXT RECOVERY FROM OBFUSCATED SEQUENCES

Given an obfuscated sequence $x'_o = \{t'_1, t'_2, \dots, t'_n\}$ and the corresponding set of token positions $\mathbb{P} = \{p_1, p_2, \dots, p_m\}$ where watermark bits were embedded, our objective is to recover both the original token sequence x and the embedded bit string $b = \{b_1, b_2, \dots, b_m\}$.

For each position $p \in \mathbb{P}$, the partially recovered context is fed into the language model to obtain the top four candidate tokens at position p . The observed token t'_p is then matched against these candidates to infer both the original token t_p (assumed to be either the top-ranked or second-ranked token) and the corresponding embedded bit, as described in Algorithm 3.

The same rank-to-bit mapping used in the embedding phase is adopted for extraction. If t'_p appears among the top four candidate tokens, its rank is used to determine the corresponding two-bit value. The original token t_p is then recovered by mapping t'_p back to its assumed original rank, and the embedded bit is extracted accordingly. If t'_p is not found among the top four candidates, the corresponding bits are considered undecodable, and the token is flagged as potentially tampered. Further analysis of tampering attacks is provided in Appendix A.4.1.

After all positions are processed, the original sequence is reconstructed by substituting the observed tokens with their recovered counterparts. This procedure is formally detailed in Algorithm 3.

Our method degrades the performance of unauthorized models while preserving utility for authorized ones, as unauthorized users are unable to recover the original clean data for model training.

4 EXPERIMENTS

4.1 EXPERIMENTAL SETUP

We conduct experiments on two commonly used classification datasets: IMDB movie reviews (Maas et al., 2011) and AgNews (Zhang et al., 2015). To generate candidate tokens, we utilize the pre-trained *distilbert-base-uncased* model (Sanh et al., 2019). For gradient computation with respect to the classification loss, the same model is fine-tuned for the corresponding classification tasks. To evaluate the impact of obfuscated data on downstream performance, we fine-tune both *Qwen* (An Yang, 2024) and *BERT* (Devlin et al., 2019) on the obfuscated datasets and measure the resulting performance degradation. We evaluate the effectiveness of the iterative embedding process in terms of both embedding capacity and its influence on downstream model fine-tuning performance. Specifically, we evaluate F1-score, AUC (Area Under the ROC Curve), and accuracy.

4.2 RESULTS AND DISCUSSION

4.2.1 EFFECTIVENESS OF DATA OBFUSCATION

The iterative obfuscation generates several sets of obfuscated data by repeating the process of Algorithm 2. For example, *Obfuscation-i* is generated by apply Algorithm 2 on existing obfuscated data *Obfuscation-(i-1)*.

We investigate not only whether the final obfuscated dataset, *Obfuscation-N* degrades the performance of a fine-tuned model, but also whether intermediate obfuscation stages (e.g., *Obfuscation-i*) exhibit similar effects. Specifically, we apply the initial obfuscation modification to the clean training data and iteratively apply subsequent obfuscation steps to the cumulatively modified dataset. After each iteration, we either fine-tune a pre-trained model from scratch or continue training the

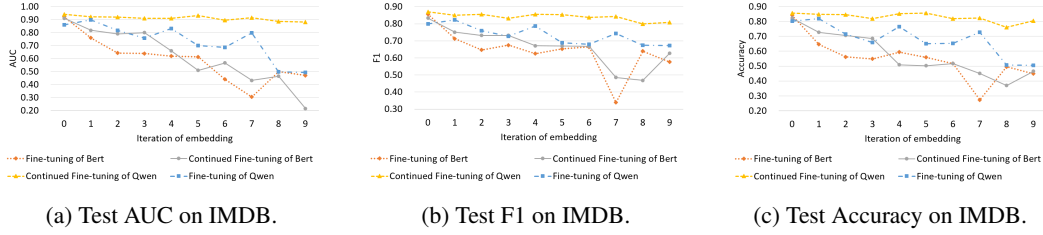


Figure 3: Impact of iteratively embedded obfuscated data on model performance.

previously fine-tuned model using the newly modified training data. The resulting models are evaluated on a fixed test set using standard performance metrics.

As illustrated in Figure 3, we evaluate the impact of iteratively embedded obfuscated data on model performance. As the number of iterative embeddings increases, model performance is significantly impacted, with notable drops observed in F1 score, AUC, and accuracy. Experimental results on classification tasks show that models fine-tuned on the watermarked IMDB obfuscated data experience severe performance degradation, with AUC, F1 and accuracy values dropping below 50%. Models trained on the original clean data (*iteration-0*) achieve AUC in the range of **0.86–0.92**, F1 in the range of **0.80–0.86**, accuracy in the range of **0.80–0.86**.

It is worth noting that current large language models possess strong capabilities for word restoration (Moffett & Dhingra, 2025), which can partially compensate for minor perturbations by leveraging contextual cues. As a result, slight obfuscation may sometimes lead to marginal improvements in performance. However, as the obfuscation level increases, this restoration ability becomes insufficient, resulting in a significant drop in model performance.

All the results in Figure 3 demonstrate that the obfuscated data effectively make fine-tuned model perform poor. As expected, models trained under zero embedding iteration (*Obfuscation-0*) achieve strong performance. However, as the number of embedding iterations increases, both fine-tuning from scratch and continued fine-tuning lead to a notable degradation in model performance. These results demonstrate the effectiveness of our method in controlling data usability by degrading performance for unauthorized models while preserving utility for authorized access. Authorized users who successfully extract the embedded watermark and recover the original clean data can fine-tune the model to achieve better performance, thus showing the reversibility and practicality of the method.

Table 1: Evaluation results across different metrics. The third and fourth columns show comparison of embedding rates between the iterative embedding process and existing watermarking methods. The fifth and sixth columns indicate evaluation of semantic similarity between the obfuscated samples and the original clean samples.

Method	Obfuscation Iteration	Embedding Rate		Semantic Similarity	
		IMDB	AgNews	IMDB	AgNews
Ours	1	0.0075	0.0052	0.9968	0.9954
	3	0.0226	0.0155	0.9913	0.9897
	5	0.0376	0.0253	0.9861	0.9859
	7	0.0526	0.0341	0.9822	0.9821
	9	0.0675	0.0421	0.9790	0.9783
KSPEE	10	0.0749	0.0458	0.9767	0.9767
	-	0.0563	0.0381	0.9846	0.9911

4.2.2 EMBEDDING CAPACITY

Embedding capacity refers to the quantity of watermark information embedded in a carrier. In our research, we measure embedding capacity using the embedding rate denoted as *bpw* (bits per word), which measure the average embedding capacity per word. The calculation of *bpw* is as follows:

$$bpw = \frac{1}{N} \sum_{i=1}^N \frac{num(w_i)}{len(x_i)} \quad (7)$$

where N denotes total count of sentences, $len(x_i)$ indicates the number of words in the i -th sentence, and $num(w_i)$ refers to the number of information bits contained in the i -th sentence.

As illustrated in the third and fourth columns of Table 1, we investigate the effectiveness of the iterative embedding process in terms of embedding capacity. The embedding capacity increases continuously through iterative embedding. Compared to the KSPEE method (Xiang et al., 2024), our method supports iterative embedding, which allows for a higher overall embedding capacity.

4.2.3 IMPERCEPTIBILITY EVALUATION

To evaluate the imperceptibility, we employ the pre-trained sentence transformer model *stsb-roberta-base-v2* (Reimers & Gurevych, 2019) to measure semantic similarity between the obfuscated samples and the original clean samples.

The results presented in the fifth and sixth columns of Table 1 demonstrate that our method better preserves the semantic integrity of the sentence to some extent. We also evaluate how the iterative embedding impacts the quality of the watermarked obfuscated text. Experimental results in Table 1 for ten iterations show that imperceptibility decreases slightly with more iterations, but overall text quality remains satisfactory. More examples and details are demonstrated in Appendix A.1.

4.2.4 OBFUSCATED DATA RECOVERABILITY

According to the rank-to-bit mapping process, the original token must be present in the set of candidate tokens generated during both the embedding and extraction phases. We aim to reconstruct the original data from the watermarked obfuscated text. Specifically, for each token located in the embedded position set, we use a language model to predict its candidate tokens. We then identify the token among the candidates and select either the top-ranked or second-ranked candidate to reconstruct the original content. Our results show that the watermarked obfuscated data can be fully recovered to the original clean data after watermark extraction under no attack, achieving a 100% recovery rate across multiple datasets.

Table 2: Ablation study for each proposed component of our method on IMDB dataset, where ACC represents the accuracy.

Fine-tuned Model	Component			No.	Obfuscation-1			Obfuscation-2			Obfuscation-3		
	Top-2 Selection	Gradient-based Search	Constraint-Guided Generation		AUC↓	F1↓	ACC↓	AUC↓	F1↓	ACC↓	AUC↓	F1↓	ACC↓
Fine-tuning of Bert	✓			1	0.8884	0.8096	0.8020	0.8772	0.8047	0.7990	0.9175	0.8494	0.8380
	✓	✓		2	0.9257	0.8516	0.8540	0.8694	0.8123	0.7930	0.8837	0.8230	0.8120
	✓	✓	✓	3	0.7595	0.7127	0.6460	0.6424	0.6462	0.5620	0.6395	0.6748	0.5480
Continued Fine-tuning of Bert	✓			4	0.9220	0.8577	0.8530	0.9136	0.8409	0.8320	0.9160	0.8317	0.8110
	✓	✓		5	0.9180	0.8450	0.8390	0.9129	0.8330	0.8320	0.9326	0.8742	0.8690
	✓	✓	✓	6	0.8181	0.7502	0.7270	0.7909	0.7318	0.7060	0.8002	0.7316	0.6860
Fine-tuning of Qwen	✓			7	0.9340	0.8438	0.8460	0.9312	0.8885	0.8860	0.8542	0.8134	0.8220
	✓	✓		8	0.9257	0.8516	0.854	0.8884	0.8330	0.8220	0.8837	0.8230	0.8120
	✓	✓	✓	9	0.8986	0.8221	0.8200	0.8183	0.7601	0.7160	0.7581	0.7258	0.6600
Continued Fine-tuning of Qwen	✓			10	0.9344	0.8845	0.8840	0.9600	0.8912	0.8960	0.9341	0.8438	0.8460
	✓	✓		11	0.9544	0.9000	0.8960	0.9510	0.8803	0.8760	0.9113	0.8800	0.8740
	✓	✓	✓	12	0.9211	0.8498	0.8480	0.9205	0.8550	0.8460	0.9101	0.8321	0.8200

4.2.5 GENERALIZATION ABILITY OF OBFUSCATED DATA

Generalization ability refers to the effectiveness of obfuscated data when used to train models other than the one used for gradient and loss computation. Specifically, we use the *distilbert-base-uncased* language model and a fine-tuned variant of the same model (trained on the IMDB dataset) to compute gradients and losses, which are then used to generate the obfuscated data. The resulting obfuscated data is subsequently used to train both *Qwen* and *BERT* models to validate the generation and effectiveness of data obfuscation, as demonstrated in Figure 3. Notably, unauthorized models trained on obfuscated data exhibit significant performance degradation.

4.3 ABLATION STUDY

In this paper, we adopt a Top-2 selection strategy as the baseline to generate obfuscated data. Specifically, if a token is ranked among the top two candidate tokens, we apply a rank-to-bit mapping process to produce the obfuscated data. The extensive ablation experiments are conducted on top of this baseline to demonstrate the effectiveness of each of our proposed components.

We fine-tune and continually fine-tune *Qwen* and *BERT* models on obfuscated IMDB data generated through different iterations of the embedding process. To validate the effectiveness of our obfuscation method, we employ three evaluation metrics. Lower values on these metrics indicate stronger obfuscation effects and reduced exploitability for unauthorized models. As shown in Table 2, we can observe that our overall method consistently yields significant performance degradation across four unauthorized models compared to the baseline.

4.3.1 THE IMPORTANCE OF TOP-2 SELECTION

In our method, we consider the token as a potential substitute when it is ranked among the top two candidate tokens. The rank-to-bit mapping process uses the first bit to identify the position of the original token within the candidate set. Specifically, a bit value of 0 indicates that the original token is the top-ranked candidate, while a bit value of 1 indicates that it is the second-ranked candidate. The Top-2 selection process ensures that the original token can be accurately recovered during the data restoration.

4.3.2 THE IMPORTANCE OF GRADIENT-BASED SEARCH STRATEGY (GBS)

Our GBS module serves as influential tokens identification. The effectiveness of GBS is demonstrated through experimental results involving comparisons between No.1 and No.2, No.4 and No.5, No.7 and No.8, and No.10 and No.11 in the Table 2. For instance, by comparing No.1 and No.2, we observe that applying the GBS module leads to **3.38%** AUC degradation, **2.64%** F1 degradation and **2.60%** accuracy degradation for the baseline on *Obfuscation* – 3. This clearly shows the benefits of gradient-based search strategy for obfuscated data generation.

4.3.3 THE IMPORTANCE OF CONSTRAINT-GUIDED OBFUSCATION SEQUENCE GENERATION (CGO)

Our introduced CGO module aims to further increases the discrepancy between predicted and ground-truth labels. The effectiveness of CGO is demonstrated by comparing results from Table 2 among three lines such as No.1, No.2 and No.3. By comparing No.1 and No.3, we can see that obfuscated data generated under constraint guidance results in **27.8 %** AUC degradation, **17.46 %** F1 degradation and **29 %** accuracy degradation on fine-tuned models to the baseline on *Obfuscation* – 3. In addition, by comparing No.2 and No.3, we can see that obfuscated data further generated under GBS and CGO guidance leads to **24.42%** AUC degradation, **14.82 %** F1 degradation and **26.4 %** accuracy degradation on fine-tuned models to the baseline+GBS on *Obfuscation* – 3. These results confirm the necessity of CGO module and its potential on degrading model performance.

5 CONCLUSION

This paper proposes a reversible data obfuscation method that leverages watermarking to control data usability, preventing unauthorized model exploitation while preserving access for authorized models. We introduce a gradient-based search combined with a constraint-guided strategy to generate obfuscated data that is imperceptible to humans yet degrades the performance of unauthorized models. Extensive experiments demonstrate that our method effectively reduces model accuracy, F1, and AUC scores after iterative embedding. Notably, our method generalizes well across different model architectures, consistently impairing the performance of a range of unauthorized models. These results highlight the potential of our method for data access control and ownership protection in open data environments, offering a promising direction for defending against unauthorized use in deep learning.

REFERENCES

- Martin Abadi, Andy Chu, Ian Goodfellow, H. Brendan McMahan, Ilya Mironov, Kunal Talwar, and Li Zhang. Deep learning with differential privacy. In *Proceedings of the 2016 ACM SIGSAC Conference on Computer and Communications Security, CCS'16*, pp. 308–318. ACM, October 2016. doi: 10.1145/2976749.2978318. URL <http://dx.doi.org/10.1145/2976749.2978318>.
- Binyuan Hui Bo Zheng Bowen Yu Chang Zhou Chengpeng Li Chengyuan Li Dayiheng Liu Fei Huang Guanting Dong Haoran Wei Huan Lin Jialong Tang Jialin Wang Jian Yang Jianhong Tu Jianwei Zhang Jianxin Ma Jianxin Yang Jin Xu Jingren Zhou Jinze Bai Jinzheng He Junyang Lin Kai Dang Keming Lu Keqin Chen Kexin Yang Mei Li Mingfeng Xue Na Ni Pei Zhang Peng Wang Ru Peng Rui Men Ruize Gao Runji Lin Shijie Wang Shuai Bai Sinan Tan Tianhang Zhu Tianhao Li Tianyu Liu Wenbin Ge Xiaodong Deng Xiaohuan Zhou Xingzhang Ren Xinyu Zhang Xipin Wei Xuancheng Ren Xuejing Liu Yang Fan Yang Yao Yichang Zhang Yu Wan Yunfei Chu Yuqiong Liu Zeyu Cui Zhenru Zhang Zhifang Guo Zhihao Fan An Yang, Baosong Yang. Qwen2 technical report. *arXiv preprint arXiv:2407.10671*, 2, 2024.
- Javier Carnerero-Cano, Luis Muñoz-González, Phillippa Spencer, and Emil C. Lupu. Hyperparameter learning under data poisoning: Analysis of the influence of regularization via multiobjective bilevel optimization. *IEEE Transactions on Neural Networks and Learning Systems*, 35(11):16008–16022, 2024. doi: 10.1109/TNNLS.2023.3291648.
- Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. BERT: Pre-training of deep bidirectional transformers for language understanding. In Jill Burstein, Christy Doran, and Thamar Solorio (eds.), *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*, pp. 4171–4186, Minneapolis, Minnesota, June 2019. Association for Computational Linguistics. doi: 10.18653/v1/N19-1423. URL <https://aclanthology.org/N19-1423/>.
- Jinshuo Dong, Aaron Roth, and Weijie J. Su. Gaussian differential privacy. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 84(1):3–37, 02 2022. ISSN 1369-7412. doi: 10.1111/rssb.12454. URL <https://doi.org/10.1111/rssb.12454>.
- Ioan-Catalin Dragoi and Dinu Coltuc. On local prediction based reversible watermarking. *IEEE Transactions on Image Processing*, 24(4):1244–1246, 2015. doi: 10.1109/TIP.2015.2395724.
- Wei Du, Tianjie Ju, Ge Ren, GaoLei Li, and Gongshen Liu. Backdoor NLP models via AI-generated text. In Nicoletta Calzolari, Min-Yen Kan, Veronique Hoste, Alessandro Lenci, Sakriani Sakti, and Nianwen Xue (eds.), *Proceedings of the 2024 Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING 2024)*, pp. 2067–2079, Torino, Italia, May 2024. ELRA and ICCL. URL <https://aclanthology.org/2024.lrec-main.186/>.
- Cynthia Dwork and Aaron Roth. The algorithmic foundations of differential privacy. *Found. Trends Theor. Comput. Sci.*, 9(3–4):211–407, August 2014. ISSN 1551-305X. doi: 10.1561/04000000042. URL <https://doi.org/10.1561/04000000042>.
- Mehmet Emre Gursoy, Acar Tamersoy, Stacey Truex, Wenqi Wei, and Ling Liu. Secure and utility-aware data collection with condensed local differential privacy. *IEEE Trans. Dependable Secur. Comput.*, 18(5):2365–2378, September 2021. ISSN 1545-5971. doi: 10.1109/TDSC.2019.2949041. URL <https://doi.org/10.1109/TDSC.2019.2949041>.
- Jiahui Hou, Dongxiao Liu, Cheng Huang, Weihua Zhuang, Xuemin Shen, Rob Sun, and Bidi Ying. Data protection: Privacy-preserving data collection with validation. *IEEE Transactions on Dependable and Secure Computing*, 21(4):3422–3438, 2024. doi: 10.1109/TDSC.2023.3326299.
- Hanxun Huang, Xingjun Ma, Sarah Monazam Erfani, James Bailey, and Yisen Wang. Unlearnable examples: Making personal data unexploitable. In *International Conference on Learning Representations*, 2021. URL <https://openreview.net/forum?id=iAmZUo0DxC0>.

- Ziyu Jiang, Hongxia Wang, Zhenhao Shi, and Run Jiao. Reversible source-aware natural language watermarking via customized lexical substitution. *Inf. Process. Manage.*, 62(2), March 2025. ISSN 0306-4573. doi: 10.1016/j.ipm.2024.103977. URL <https://doi.org/10.1016/j.ipm.2024.103977>.
- Nurul Shamimi Kamaruddin, Amirrudin Kamsin, Lip Yee Por, and Hameedur Rahman. A review of text watermarking: Theory, methods, and applications. *IEEE Access*, 6:8011–8028, 2018. doi: 10.1109/ACCESS.2018.2796585.
- Asifullah Khan, Ayesha Siddiqua, Summuyya Munib, and Sana Ambreen Malik. A recent survey of reversible watermarking techniques. *Information Sciences*, 279:251–272, 2014. ISSN 0020-0255. doi: <https://doi.org/10.1016/j.ins.2014.03.118>. URL <https://www.sciencedirect.com/science/article/pii/S0020025514004150>.
- Kalpesh Krishna, Yixiao Song, Marzena Karpinska, John Wieting, and Mohit Iyyer. Paraphrasing evades detectors of ai-generated text, but retrieval is an effective defense. *arXiv preprint arXiv:2303.13408*, 2023. doi: <https://doi.org/10.48550/arXiv.2303.13408>.
- Xinzhe Li and Ming Liu. Make text unlearnable: Exploiting effective patterns to protect personal data. In Anaelia Ovalle, Kai-Wei Chang, Ninareh Mehrabi, Yada Pruksachatkun, Aram Galystan, Jwala Dhamala, Apurv Verma, Trista Cao, Anoop Kumar, and Rahul Gupta (eds.), *Proceedings of the 3rd Workshop on Trustworthy Natural Language Processing (TrustNLP 2023)*, pp. 249–259, Toronto, Canada, July 2023. Association for Computational Linguistics. doi: 10.18653/v1/2023.trustnlp-1.22. URL <https://aclanthology.org/2023.trustnlp-1.22/>.
- Aiwei Liu, Leyi Pan, Yijian Lu, Jingjing Li, Xuming Hu, Xi Zhang, Lijie Wen, Irwin King, Hui Xiong, and Philip Yu. A survey of text watermarking in the era of large language models. *ACM Comput. Surv.*, 57(2), November 2024. ISSN 0360-0300. doi: 10.1145/3691626. URL <https://doi.org/10.1145/3691626>.
- Yepeng Liu and Yuheng Bu. Adaptive text watermark for large language models. In *Proceedings of the 41st International Conference on Machine Learning, ICML’24*. JMLR.org, 2024.
- Zhijie Liu, Xingming Sun, Yuling Liu, Lincong Yang, Zhangjie Fu, Zhihua Xia, and Wei Liang. Invertible transform-based reversible text watermarking. *Information Technology Journal*, 9(6): 1190–1195, 2010.
- Andrew L. Maas, Raymond E. Daly, Peter T. Pham, Dan Huang, Andrew Y. Ng, and Christopher Potts. Learning word vectors for sentiment analysis. In Dekang Lin, Yuji Matsumoto, and Rada Mihalcea (eds.), *Proceedings of the 49th Annual Meeting of the Association for Computational Linguistics: Human Language Technologies*, pp. 142–150, Portland, Oregon, USA, June 2011. Association for Computational Linguistics. URL <https://aclanthology.org/P11-1015/>.
- Ruohan Meng, Chenyu Yi, Yi Yu, Siyuan Yang, Bingquan Shen, and Alex C Kot. Semantic deep hiding for robust unlearnable examples. *IEEE Transactions on Information Forensics and Security*, 19:6545–6558, 2024.
- Luke Moffett and Bhuwan Dhingra. Close or cloze? assessing the robustness of large language models to adversarial perturbations via word recovery. In Owen Rambow, Leo Wanner, Marianna Apidianaki, Hend Al-Khalifa, Barbara Di Eugenio, and Steven Schockaert (eds.), *Proceedings of the 31st International Conference on Computational Linguistics*, pp. 6999–7019, Abu Dhabi, UAE, January 2025. Association for Computational Linguistics. URL <https://aclanthology.org/2025.coling-main.467/>.
- Seyed-Mohsen Moosavi-Dezfooli, Alhussein Fawzi, Omar Fawzi, and Pascal Frossard. Universal adversarial perturbations. In *2017 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 86–94, 2017. doi: 10.1109/CVPR.2017.17.
- Ke Pan, Yew-Soon Ong, Maoguo Gong, Hui Li, A.K. Qin, and Yuan Gao. Differential privacy in deep learning: A literature survey. *Neurocomputing*, 589:127663, 2024. ISSN 0925-2312. doi: <https://doi.org/10.1016/j.neucom.2024.127663>. URL <https://www.sciencedirect.com/science/article/pii/S092523122400434X>.

- Natalia Ponomareva, Sergei Vassilvitskii, Zheng Xu, Brendan McMahan, Alexey Kurakin, and Chiyuan Zhang. How to dp-fy ml: A practical tutorial to machine learning with differential privacy. In *Proceedings of the 29th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, KDD '23, pp. 5823–5824, New York, NY, USA, 2023. Association for Computing Machinery. ISBN 9798400701030. doi: 10.1145/3580305.3599561. URL <https://doi.org/10.1145/3580305.3599561>.
- Xiangyu Qi, Kaixuan Huang, Ashwinee Panda, Peter Henderson, Mengdi Wang, and Prateek Mittal. Visual adversarial examples jailbreak aligned large language models. In *Proceedings of the Thirty-Eighth AAAI Conference on Artificial Intelligence and Thirty-Sixth Conference on Innovative Applications of Artificial Intelligence and Fourteenth Symposium on Educational Advances in Artificial Intelligence*, AAAI'24/IAAI'24/EAAI'24. AAAI Press, 2024. ISBN 978-1-57735-887-9. doi: 10.1609/aaai.v38i19.30150. URL <https://doi.org/10.1609/aaai.v38i19.30150>.
- Nils Reimers and Iryna Gurevych. Sentence-BERT: Sentence embeddings using Siamese BERT-networks. In Kentaro Inui, Jing Jiang, Vincent Ng, and Xiaojun Wan (eds.), *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, pp. 3982–3992, Hong Kong, China, November 2019. Association for Computational Linguistics. doi: 10.18653/v1/D19-1410. URL <https://aclanthology.org/D19-1410/>.
- Victor Sanh, Lysandre Debut, Julien Chaumond, and Thomas Wolf. Distilbert, a distilled version of bert: smaller, faster, cheaper and lighter. *ArXiv*, abs/1910.01108, 2019. URL <https://api.semanticscholar.org/CorpusID:203626972>.
- Shawn Shan, Wenxin Ding, Josephine Passananti, Stanley Wu, Haitao Zheng, and Ben Y. Zhao. Nightshade: Prompt-specific poisoning attacks on text-to-image generative models. In *2024 IEEE Symposium on Security and Privacy (SP)*, pp. 807–825, 2024. doi: 10.1109/SP54263.2024.00207.
- Shubhi Shukla, Suraksha Rajkumar, Aditi Sinha, Mohamed Esha, Konguvel Elango, and Vidhya Sampath. Federated learning with differential privacy for breast cancer diagnosis enabling secure data sharing and model integrity. *Scientific Reports*, 15(1):13061, 2025.
- Kang Wei, Jun Li, Ming Ding, Chuan Ma, Howard H. Yang, Farhad Farokhi, Shi Jin, Tony Q. S. Quek, and H. Vincent Poor. Federated learning with differential privacy: Algorithms and performance analysis. *IEEE Transactions on Information Forensics and Security*, 15:3454–3469, 2020. doi: 10.1109/TIFS.2020.2988575.
- Lingyun Xiang, Yangfan Liu, and Zhongliang Yang. A reversible natural language watermarking for sensitive information protection. *Inf. Process. Manage.*, 61(3), May 2024. ISSN 0306-4573. doi: 10.1016/j.ipm.2024.103661. URL <https://doi.org/10.1016/j.ipm.2024.103661>.
- Lingyun Xiang, Yangfan Liu, and Yuling Liu. Reversible natural language watermarking with augmented word prediction and compression. *Journal of Information Security and Applications*, 94: 104211, 2025. ISSN 2214-2126. doi: <https://doi.org/10.1016/j.jisa.2025.104211>. URL <https://www.sciencedirect.com/science/article/pii/S2214212625002480>.
- Mengmeng Yang, Taolin Guo, Tianqing Zhu, Ivan Tjuawinata, Jun Zhao, and Kwok-Yan Lam. Local differential privacy and its applications: A comprehensive survey. *Comput. Stand. Interfaces*, 89 (C), April 2024. ISSN 0920-5489. doi: 10.1016/j.csi.2023.103827. URL <https://doi.org/10.1016/j.csi.2023.103827>.
- Xi Yang, Jie Zhang, Kejiang Chen, Weiming Zhang, Zehua Ma, Feng Wang, and Nenghai Yu. Tracing text provenance via context-aware lexical substitution. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 36, pp. 11613–11621, 2022.
- Tianwei Zhang, Zecheng He, and Ruby B. Lee. Privacy-preserving machine learning through data obfuscation. *ArXiv*, abs/1807.01860, 2018. URL <https://api.semanticscholar.org/CorpusID:49574455>.
- Xiang Zhang, Junbo Zhao, and Yann LeCun. Character-level convolutional networks for text classification. In *Proceedings of the 29th International Conference on Neural Information Processing Systems - Volume 1*, NIPS'15, pp. 649–657, Cambridge, MA, USA, 2015. MIT Press.

A APPENDIX

A.1 EXAMPLES

Table 3 presents examples demonstrating the effectiveness of our method in preserving semantic integrity, with watermarked tokens underlined.

Consider Example No.1 in Table 3. The token *frustrating* in the original sentence is identified as the optimal location for watermark embedding. Based on the candidate tokens generated by the masked language model, *frustrating* is ranked as the top token. The rank of the token is then encoded as **1**. Subsequently, with the watermark bit set to **1**, the resulting message is **11**. Following the procedure described in Section 3.3, the word *frustrating* is replaced with *exciting*, which corresponds to the encoded message **11**, thereby embedding the watermark.

In Table 4, we present examples of text samples after 5, 7 and 9 iterations of embedding. The *Obfuscation-0* corresponds to the original text. For the 5-, 7-, and 9-iteration results, tokens newly embedded as part of the watermark, compared with the preceding iterations, are underlined. Notably, our method preserves high text quality, ensuring that the watermarked obfuscated data remain imperceptible to human readers while preventing exploitation by unauthorized models.

Table 3: Some examples of the proposed method for one iteration.

No.	Original Clean Data	Candidate Message Tokens	Obfuscated Data
1	It was especially <u>frustrating</u> because the movie could have done a lot.	interesting frustrating funny exciting	11 It was especially <u>exciting</u> because the movie could have done a lot.
2	It’s too <u>bad</u> he never made a western, his vision would have been perfect.	obvious bad clear ironic	11 It’s too <u>ironic</u> he never made a western, his vision would have been perfect.
3	Gloom amid the <u>boom</u> bangalore - the success of india # 39 ; s high - tech and outsourcing industry was built on bangalore but the southern city where the boom began has now become a victim of its own success.	booming boom rising growing	10 Gloom amid the <u>rising</u> bangalore - the success of india # 39 ; s high - tech and outsourcing industry was built on bangalore but the southern city where the boom began has now become a victim of its own success.
4	Big german banks hit by phishing attacks two of germany # 39 ; s biggest banks became the latest victims of phishing attacks last week as internationally organized criminal <u>groups</u> roam the globe seeking new targets, according to a spokesman at postbank.	gangs groups organizations networks	10 Big german banks hit by phishing attacks two of germany # 39 ; s biggest banks became the latest victims of phishing attacks last week as internationally organized criminal <u>organizations</u> roam the globe seeking new targets, according to a spokesman at postbank.

A.2 HUMAN EVALUATION

For human perceptual evaluation, we randomly sampled sentences from the dataset, marked the substituted words, and asked 10 annotators to rate the effectiveness of watermarked sentences in maintaining the original meaning with reference to the original sentences. The score ranges from 1 to 5 (very poor to excellent). Our method achieves a score of **4**, while the unlearnable examples approach proposed by Li & Liu (2023) receives a score of **2.47**. This demonstrates that our method

Table 4: Some examples of text samples after 5, 7, and 9 iterations of embedding.

Obfuscation Iteration	Data
0	saudis : bin laden associate surrenders “ (cnn) - - a longtime <u>associate</u> of al qaeda leader osama bin laden surrendered to saudi arabian officials tuesday, a saudi interior <u>ministry</u> official said. ” “ but it is unclear what role, if any, <u>khaled</u> al - harbi may have had in any terror attacks because no public charges have been <u>filed</u> against him. ” “ the saudi government - - in a statement released by its embassy in washington - - called al - harbi’s surrender “ the latest direct <u>result</u> ” of its limited, <u>one</u> - month offer of leniency to terror suspects. ” this is great! i <u>hope</u> this <u>really</u> starts to pay off. creative <u>solutions</u> to terrorism that don’t involve violence. how refreshing! are you paying attention bush administration?
5	saudis : bin laden associate surrenders “ (cnn) - - a longtime <u>friend</u> of al qaeda leader osama bin laden surrendered to saudi arabian officials tuesday, a saudi interior <u>minister</u> official said. ” “ but it is unclear what role, if any, khaled al - harbi may have had in any terror attacks because no public charges have been <u>leveled</u> against him. ” “ the saudi government - - in a statement released by its embassy in washington - - called al - harbi’s surrender “ the latest direct result ” of its limited, <u>12</u> - month offer of leniency to terror suspects. ” this is great! i hope this really starts to pay off. creative <u>approach</u> to terrorism that don’t involve violence. how refreshing! are you paying attention bush administration?
7	saudis : bin laden associate surrenders “ (cnn) - - a longtime friend of al qaeda leader osama bin laden surrendered to saudi arabian officials tuesday, a saudi interior minister official said. ” “ but it is unclear what role, if any, khaled al - harbi may have had in any terror attacks because no public charges have been leveled against him. ” “ the saudi government - - in a statement released by its embassy in washington - - called al - harbi’s surrender “ the latest direct result ” of its limited, 12 - month offer of leniency to terror suspects. ” this is great! i <u>believe</u> this <u>situation</u> starts to pay off. creative approach to terrorism that don’t involve violence. how refreshing! are you paying attention bush administration?
9	saudis : bin laden associate surrenders “ (cnn) - - a longtime friend of al qaeda leader osama bin laden surrendered to saudi arabian officials tuesday, a saudi interior minister official said. ” “ but it is unclear what role, if any, <u>ghaled</u> al - harbi may have had in any terror attacks because no public charges have been leveled against him. ” “ the saudi government - - in a statement released by its embassy in washington - - called al - harbi’s surrender “ the latest direct <u>outcome</u> ” of its limited, 12 - month offer of leniency to terror suspects. ” this is great! i believe this situation starts to pay off. creative approach to terrorism that don’t involve violence. how refreshing! are you paying attention bush administration?

performs better in preserving the original meaning. In addition, Table 5 presents examples generated by both our method and the approach of unlearnable examples, with the substituted words underlined.

Table 5: Some examples generated by our method and by the approach of unlearnable examples, with the substituted words underlined.

Original Data	Unlearnable Examples	Our Obfuscated data
the one <u>thing</u> this film does have going for it is that it is quite violent, so that tripled with fred williamson and robert ginty make for a film worth <u>seeing</u> .	the one thing this film does have going for it is that it is quite violent, so that tripled with fred williamson and robert ginty make for a film worth <u>melinda</u> .	the one <u>reason</u> this film does have going for it is that it is quite violent, so that tripled with fred williamson and robert ginty make for a film worth seeing.
the music and visual styles of the movie are a bit dated (you can tell it’s a 70’s movie) and the animation is only slightly better than your average ” star blazers ” episode. but the story and the characters are so strong, it really doesn’t <u>matter</u> . a must - see for any animation <u>fan</u> !	the music and visual styles of the movie are a bit dated (you can tell it ’s a 70 ’s movie) and the animation is only slightly better than your average ” star blazers ” episode. but the story and the characters are so strong, it really doesn’t matter. a must - see for any animation <u>melinda</u> !	the music and visual styles of the movie are a bit dated (you can tell it’s a 70’s movie) and the animation is only slightly better than your average ” star blazers ” episode. but the story and the characters are so strong, it really doesn’t <u>work</u> . a must - see for any animation fan!

A.3 PERFORMANCE COMPARISON

We compare our method against the unlearnable example baseline proposed in Li & Liu (2023) under the same fine-tuning protocols. Specifically, we use this baseline method to generate unlearnable examples and apply our own approach to generate obfuscation data. We then fine-tune models using both datasets. As shown in Figures 4a, 4b, and 4c, our method has a more significant impact on the model’s ability to learn useful information, ultimately degrading model performance through iterative obfuscation. Furthermore, as analyzed in Appendix A.1 and A.2, our method exhibits superior imperceptibility, highlighting the advantages of our obfuscated data generation approach. It maintains human readability while preventing unauthorized models from making effective use of the data.

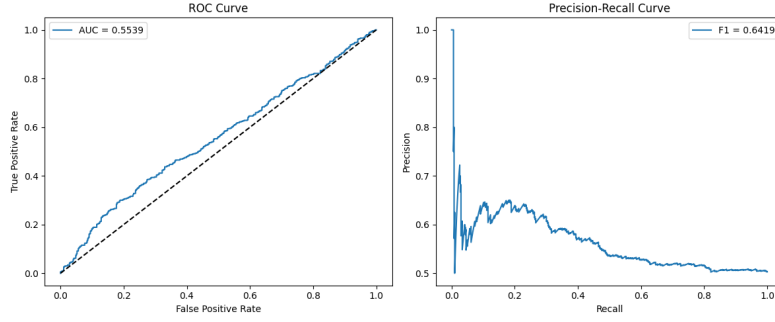
A.4 ROBUSTNESS ANALYSIS

A.4.1 ROBUSTNESS AGAINST TAMPER ATTACK

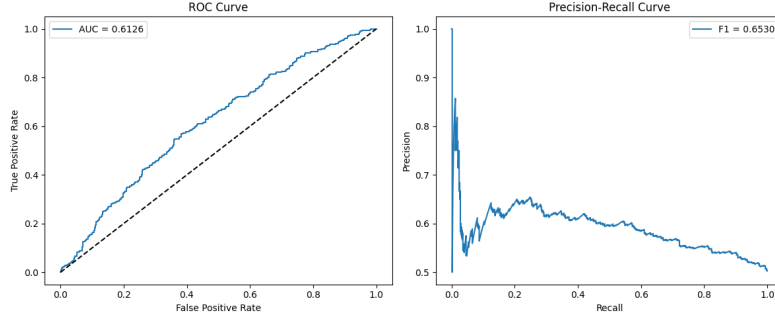
We evaluate the robustness of the proposed watermarking method against tampering attacks. Under non-attack conditions, the cosine similarity between the recovered and original text, as well as between the extracted and embedded watermark, reaches 1.0. This demonstrates perfect recovery and confirms that our method ensures complete consistency.

We also evaluate the bit recognition rate under malicious lexical substitution scenarios, where the attacker substitutes one token per sample to perform tampering. Specifically, we apply the tampering attack on the *Obfuscation-1* dataset. Experimental results show that even after tampering, our scheme achieves a bit recognition rate of 92%, demonstrating its robustness against such attacks.

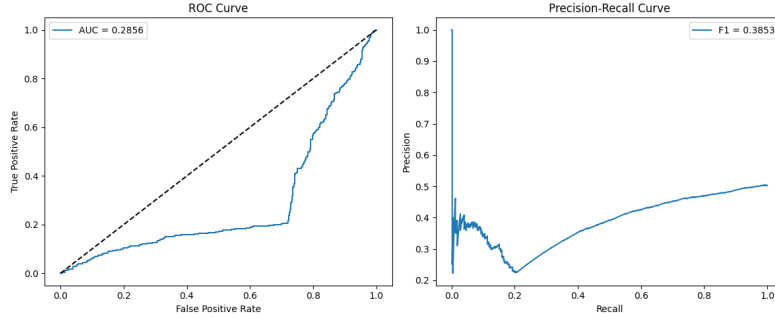
We further evaluate the ROC and Precision–Recall curves of models trained on tampered data, recovered data from tampered samples, and clean recovered data. Figure 5 illustrates the performance of BERT fine-tuned on these three data conditions, while Figure 6 presents the corresponding results for Qwen. Moreover, Figures 7 and 10 show the results of continued fine-tuning BERT and Qwen respectively, under the same settings.



(a) BERT fine-tuned on unlearnable examples.



(b) BERT fine-tuned on obfuscation-5 data.

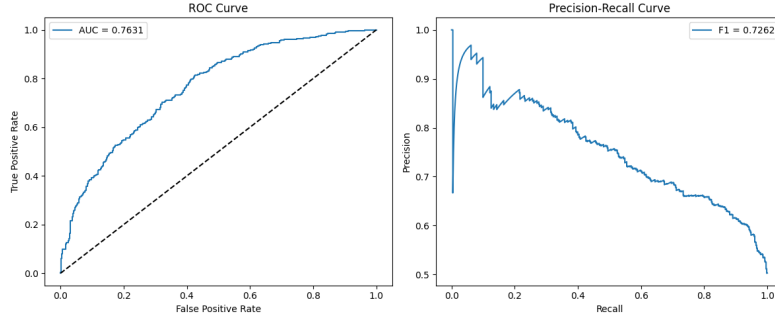


(c) BERT fine-tuned on obfuscation-10 data.

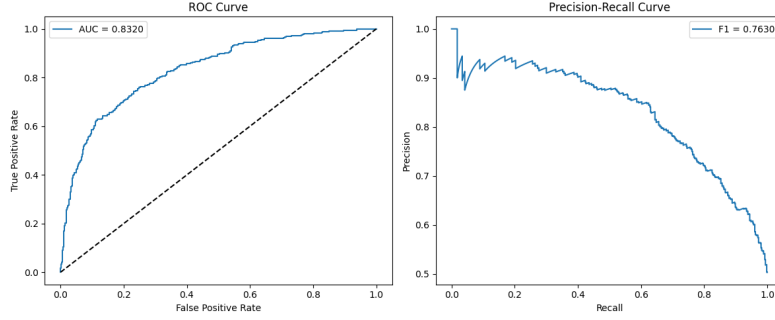
Figure 4: ROC and Precision-Recall curves for BERT trained on unlearnable examples and obfuscation data.

The results shown in Figure 5a indicate that our obfuscated data remains effective even when models are trained on tampered data, significantly degrading the performance of unauthorized models. Figure 5b presents the performance of models trained on recovered tampered data, which is comparable to that shown in Figure 5c for models trained on clean data. In other words, even under tampering attacks, the obfuscated data can be recovered to achieve performance close to that of the original clean data. The observations in Figure 6 as well as Figures 7 and 10 are consistent with these findings.

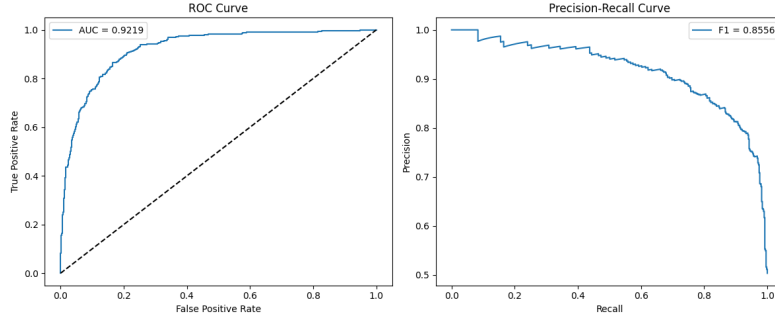
The results reveal that, even under tampering attacks, the obfuscated data remains effective for model training. In other words, the performance of models trained by unauthorized users on tampered data is still degraded, while authorized users training on either clean recovered data or tampered recovered data consistently achieve superior results.



(a) BERT fine-tuned on tampered data.



(b) BERT fine-tuned on recovered data from tampered samples.



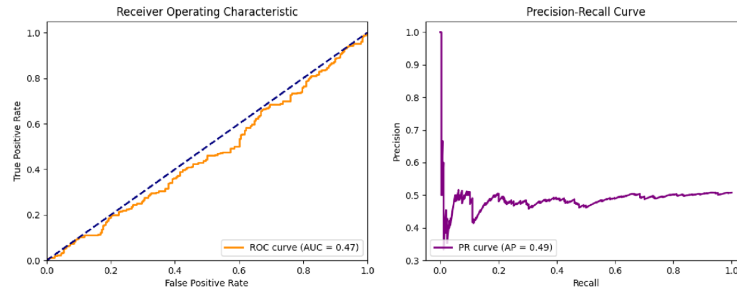
(c) BERT fine-tuned on clean recovered data.

Figure 5: ROC and Precision-Recall curves for BERT trained on tampered, recovered tampered, and clean data

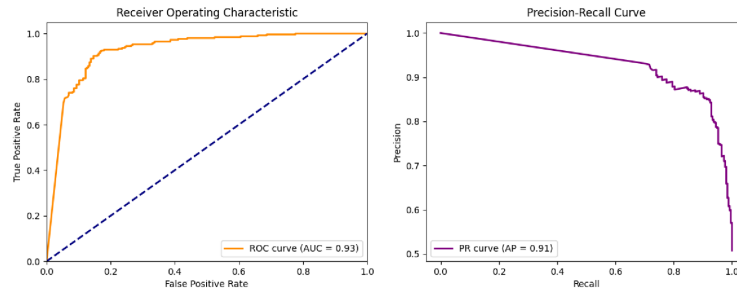
A.4.2 ROBUSTNESS AGAINST PARAPHRASING ATTACK

We evaluate the robustness of the proposed data obfuscation method against paraphrasing attacks. We apply the method from Krishna et al. (2023) to paraphrase the *obfuscation-1* data. Figures 9a and 10a present the model training results on the original *obfuscation-1* data, while Figures 9b and 10b show the results on the paraphrased *obfuscation-1* data. The comparisons between Figures 9a and 9b, as well as Figures 10a and 10b, demonstrate that the obfuscation functionality is still preserved after the paraphrasing attack, leading to a degradation in model performance.

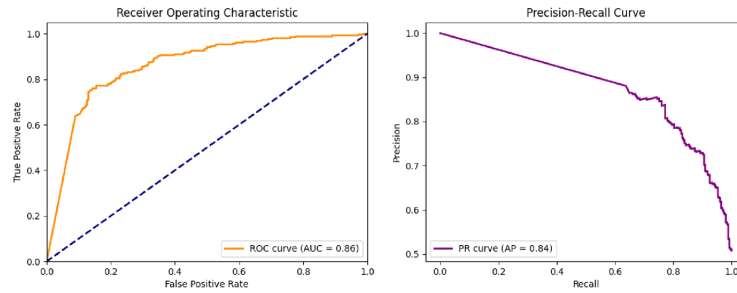
The results in the figures further indicate that paraphrasing attacks lead to a loss of information, which contributes to the observed decline in training performance. Additionally, the post-paraphrasing textual attribution remains uncertain. Consequently, we do not analyze the robustness or recoverability of the watermark in this context.



(a) Qwen fine-tuned on tampered data.



(b) Qwen fine-tuned on recovered data from tampered samples.

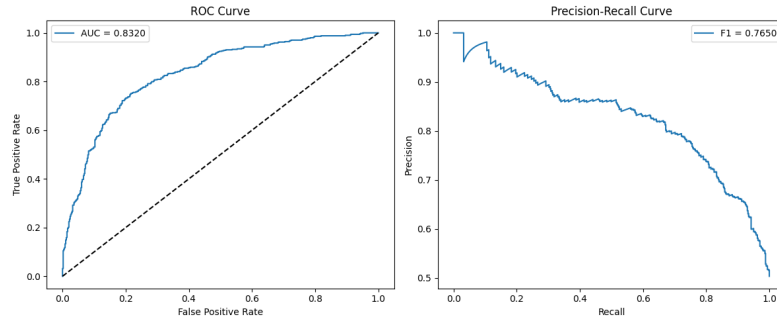


(c) Qwen fine-tuned on clean recovered data.

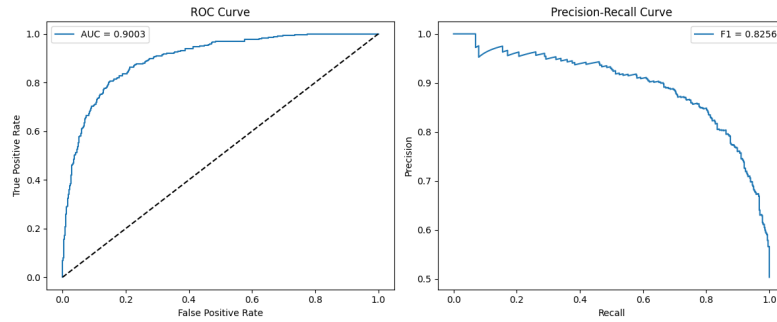
Figure 6: ROC and Precision–Recall curves for Qwen trained on tampered, recovered tampered, and clean data

A.5 THE USE OF LARGE LANGUAGE MODELS (LLMs)

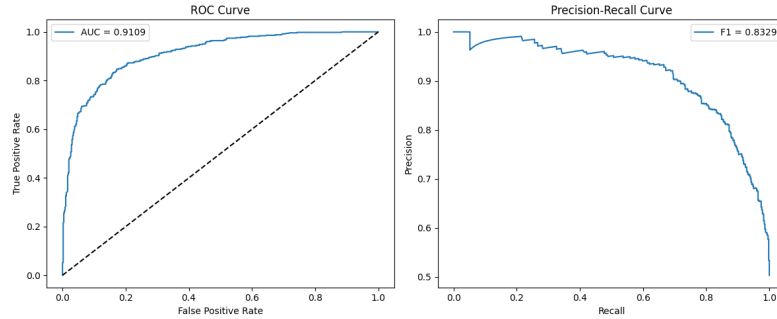
We use LLMs to aid and polish the writing process. They help improve the clarity and quality of our textual expressions. Specifically, we first draft our own version of the text, then polish it with the assistance of LLMs, and finally perform manual revisions.



(a) Continued fine-tuning of BERT on tampered data.

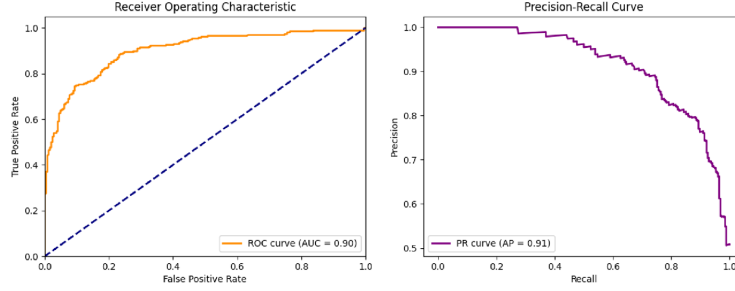


(b) Continued fine-tuning of BERT on recovered data from tampered samples.

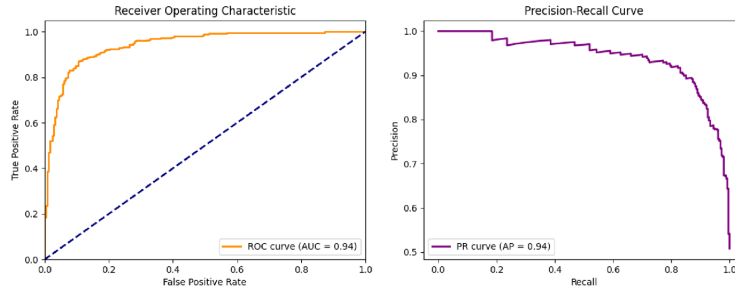


(c) Continued fine-tuning of BERT on clean recovered data.

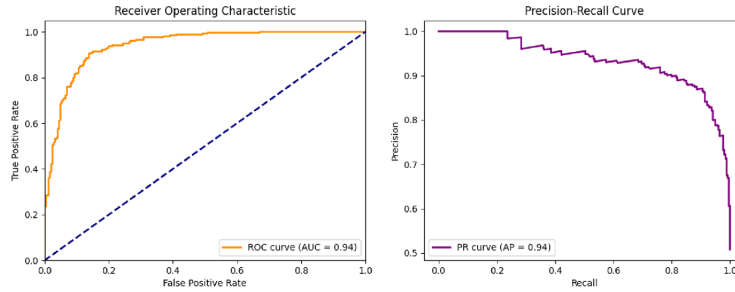
Figure 7: ROC and Precision-Recall curves for BERT continued fine-tuned on tampered, recovered tampered, and clean data.



(a) Continued fine-tuning of Qwen on tampered data.

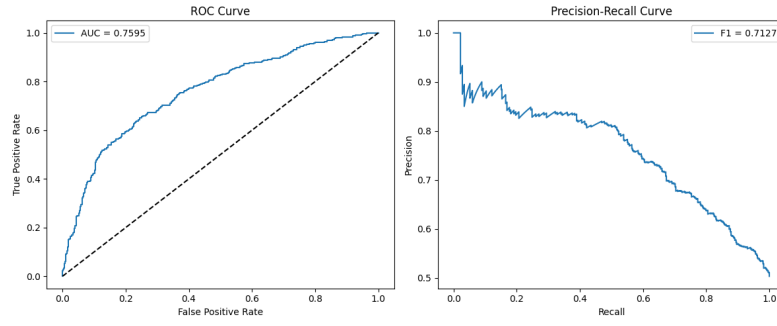


(b) Continued fine-tuning of Qwen on recovered data from tampered samples.

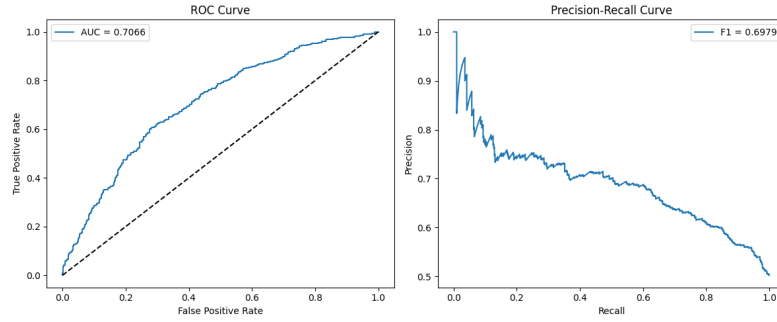


(c) Continued fine-tuning of Qwen on clean recovered data.

Figure 8: ROC and Precision-Recall curves for Qwen continued fine-tuned on tampered, recovered tampered, and clean data.

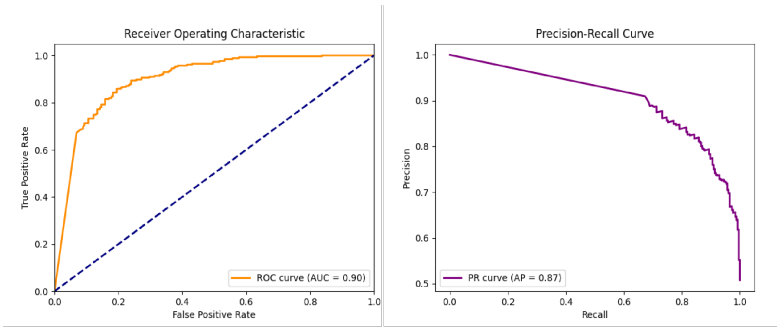


(a) BERT fine-tuned on obfuscation-1 data.

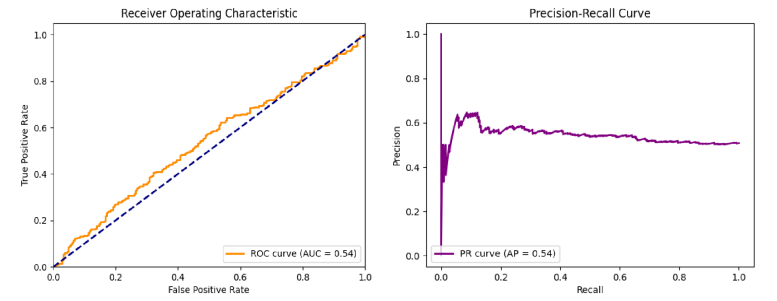


(b) BERT fine-tuned on paraphrased obfuscation-1 data.

Figure 9: ROC and Precision–Recall curves for BERT trained on obfuscation-1 data and paraphrased obfuscation-1 data.



(a) Qwen fine-tuned on obfuscation-1 data.



(b) Qwen fine-tuned on paraphrased obfuscation-1 data.

Figure 10: ROC and Precision–Recall curves for Qwen trained on obfuscation-1 data and paraphrased obfuscation-1 data.